



HINDUSTAN
INSTITUTE OF TECHNOLOGY & SCIENCE
(DEEMED TO BE UNIVERSITY)

DEPARTMENT OF MECHATRONICS ENGINEERING

INNOVATION LAB PROJECT REPORT

AUTO IRRIGATION SYSTEM

EMD51401 DESIGN AND PROTOTYPING

SUBMITTED BY:

TEAM MEMBER NAME & REG NO.

1.Thirisha.R

[22125022]

BONAFIDE CERTIFICATE

Name of the course : **DESIGN AND PROTOTYPING**

Department: **MECHATRONICS**

Certified that this a Bonafide record of the work done by

Thirisha.R [22125022]

of **MECHATRONICS** Class in the **Design And Prototyping**

during the year **2022-2023**.

NAME OF THE EXAMINATION :

DESIGN AND PROTOTYPING

Introduction:

The concept of auto irrigation system is gaining popularity due to its efficiency and ease of use. The system helps in automating the process of watering plants and crops, based on the moisture level in the soil. In this report, we will discuss the components used in an auto irrigation system, including Arduino Uno R3, 5V relay, soil moisture sensor, and battery.

COMPONENTS:

1. ArduinoUno R3
2. 5v Relay
3. Soil Moisture Sensor
4. Battery

Arduino Uno R3:

Arduino Uno R3 is a microcontroller board based on the ATmega328P. It has 14 digital input/output pins, 6 analog inputs, a 16 MHz quartz crystal, a USB connection, and a power jack. The board can be programmed using the Arduino IDE, which is an open-source software platform. In an auto irrigation system, the Arduino board acts as the brain of the system, which controls the relay and soil moisture sensor.

5V Relay:

A relay is an electrically operated switch that can be controlled by a small current. The 5V relay is a type of relay that operates on 5 volts DC. It consists of a coil, which when energized, closes the contacts of the relay. The contacts can be used to switch on or off a high voltage load, such as a water pump. In an auto irrigation system, the relay is used to turn on or off the water pump based on the moisture level in the soil.

Soil Moisture Sensor:

A soil moisture sensor is an electronic device that measures the moisture level in the soil. It consists of two probes that are inserted into the soil. When the soil is dry, it has a high resistance, which decreases as the soil becomes wet. The sensor can be interfaced with the Arduino board to measure the moisture level and control the relay.

ARDUINO CODE:

```
const int relayPin = 2; // Pin connected to the relay const int
moisturePin = A0; // Pin connected to the soil moisture sensor const
int batteryPin = A1; // Pin connected to the battery voltage sensor

const int moistureThreshold = 500; // Moisture level threshold (adjust as needed)
const int batteryThreshold = 700; // Battery voltage threshold (adjust as needed)
const int pumpDuration = 5000; // Pump duration in milliseconds (adjust as
needed)

void setup() { pinMode(relayPin, OUTPUT);
digitalWrite(relayPin, LOW); // Turn off the relay by
default
Serial.begin(9600);
}

void loop() { int moistureLevel =
analogRead(moisturePin); int
batteryLevel = analogRead(batteryPin);

Serial.print("Moisture Level: ");
Serial.println(moistureLevel);
Serial.print("Battery Voltage: ");
Serial.println(batteryLevel);
```

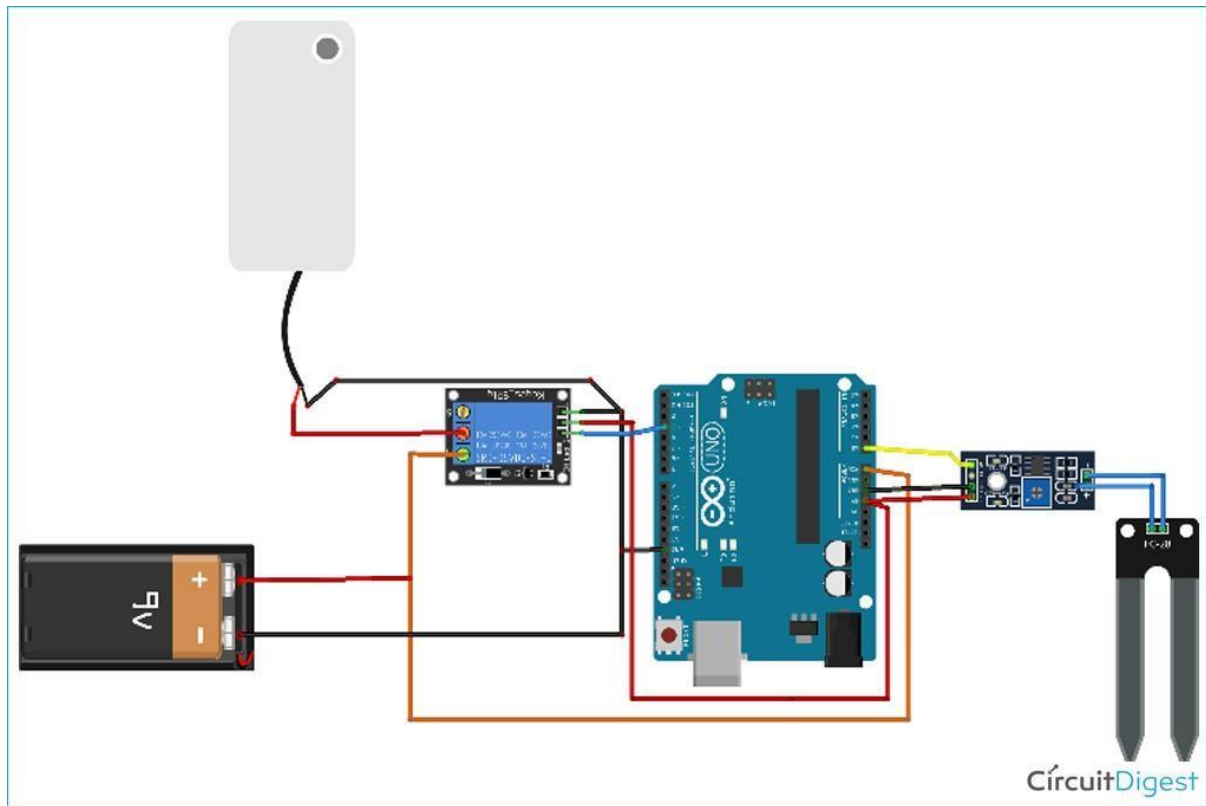
```
    if (moistureLevel > moistureThreshold && batteryLevel >
batteryThreshold) {    digitalWrite(relayPin, HIGH); // Turn on the relay
delay(pumpDuration);    digitalWrite(relayPin, LOW); // Turn off the relay
    }

    delay(1000); // Wait for 1 second before taking another reading
}
```

Conclusion:

In conclusion, an auto irrigation system is an efficient and easy-to-use solution for watering plants and crops. The system consists of several components, including Arduino Uno R3, 5V relay, soil moisture sensor, and battery. These components work together to automate the process of watering plants based on the moisture level in the soil. By using an auto irrigation system, farmers and gardeners can save time and resources while ensuring that their plants receive the right amount of water.

CIRCUIT DIAGRAM:-



BILL AMOUNT:-

1. SOIL MOISTURE SENSOR :-	90
2. RELAY MODULE 5V :-	125
3. ARDUINO UNO R3 :-	800
4. WATER PUMP :-	100
5. BATTERY 9V {2 PCS} :-	50
TOTAL	1165 RS

PROGRESS

Project Title: Automated Irrigation System

Duration: 3 months (FEBURARY 2023 - APRIL 2023)

Introduction:

The purpose of this progress report is to provide an update on the development of the Automated Irrigation System project. This project aims to create an automated system that optimizes water usage and enhances the efficiency of irrigation in agriculture. The first two months faced challenges and setbacks, but the third month showed significant progress and success.

Month 1: FEBURARY 2023

During the first month of the project, the team faced several obstacles and encountered unforeseen difficulties. The initial plan was to design and implement a system that would automatically monitor soil moisture levels and activate the irrigation process when necessary. However, the system faced technical issues, primarily related to sensor calibration and data acquisition. The team realized that the sensors used were not accurately measuring soil moisture, leading to incorrect irrigation decisions. Despite multiple attempts to recalibrate the sensors, the accuracy of measurements remained unsatisfactory.

Month 2: MARCH 2023

In the second month, the team focused on troubleshooting and addressing the issues encountered in the previous month. Various modifications and adjustments were made to the sensor setup, and alternative sensor models were tested to improve accuracy. However, these attempts did not yield satisfactory results. Additionally, time constraints and resource limitations hindered progress, making it difficult to explore alternative approaches. As a result, the second month was largely characterized by setbacks, and the project seemed to be at a standstill.

Month 3: APRIL 2023

During the third month, the team adopted a different approach to tackle the persisting challenges. After careful analysis and consultation with experts, it was decided to switch to a different type of soil moisture sensor that had proven reliability and accuracy in similar projects. The new sensor was promptly acquired and

integrated into the system. Rigorous testing and calibration were conducted to ensure accurate measurement of soil moisture levels.

Significant progress was observed with the implementation of the new sensor. The automated irrigation system successfully detected soil moisture deficits and triggered irrigation cycles accordingly. The system effectively optimized water usage by providing the right amount of irrigation at the right time, resulting in healthier plant growth and

improved crop yield. Moreover, the system's data acquisition capabilities were enhanced, allowing for more comprehensive monitoring and analysis of soil moisture levels.

Conclusion:

While the first two months of the Automated Irrigation System project were marked by challenges and setbacks, the third month witnessed significant progress and success. By replacing the faulty soil moisture sensors with a more reliable alternative, the system was able to accurately detect soil moisture deficits and optimize irrigation cycles accordingly. The successful implementation of the automated irrigation system holds promise for improving water efficiency and crop yield in agricultural settings.

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